

Multiple Choice

1. **Answer:** A

Options: A) `int(x [,base])`; B) `long(x [,base] )`; C) `float(x)`; D) `str(x)`

Note: The function `int(x [,base])` in Python 3 is specifically designed to convert a string representation of a number into its integer form, optionally allowing for a specified base for conversion.

2. **Answer:** C

Options: A) $O(N^{(1/2)})$; B) $O(N^{(1/4)})$; C) $O(N^{(1/N)})$; D) $O(N)$

Note: $O(N^{(1/N)})$ grows the slowest among functions because as N increases, the value of $N^{(1/N)}$ approaches 1, indicating that the growth rate is significantly slower than polynomial, logarithmic, or exponential functions.

3. **Answer:** D

Options: A) 1; B) 6; C) 4; D) 5

Note: The correct answer is 5 because in Python, the modulus operator (%) has a higher precedence than addition (+), so `3 % 2` is evaluated first, resulting in 1, and then `4 + 1` equals 5.

4. **Answer:** B

Options: A) a; B) Chemistry; C) 0; D) 1

Note: The correct answer is "Chemistry" because in Python lists, indexing starts at 0, so the element at index 1 of the list `['a', 'Chemistry', 0, 1]` is 'Chemistry'.

5. **Answer:** B

Options: A) 0; B) 1; C) 3; D) 4

Note: The correct answer is 1 because the expression evaluates the modulus operation first, where `3 % 3` equals 0, and then adds 1 to it, resulting in `1 + 0`, which equals 1.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Confidentiality is not compromised simply by logging into a system with valid credentials, as this action is expected and authorized, assuming the user is entitled to access the information.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the `max()` function in Python 3 is specifically designed to evaluate a list of comparable items and return the one with the highest value based on their natural ordering.

8. **Answer:** True

Options: A) True; B) False

Note: The decimal number 231 is equal to E₇ in hexadecimal because when you divide 231 by 16, the quotient is 14 (which is E in hex) and the remainder is 7, thus representing the number as E₇ in base 16.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because strings in Python are immutable, meaning they cannot be changed in place, so the 'sort()' method cannot be applied to a string.

10. **Answer:** True

Options: A) True; B) False

Note: In Python 3, the '**' operator is specifically designed to calculate the power of a number by raising the base to the exponent, making the statement true.

Long Form Response

11. **Answer:** `def divSum(n): | return sum; | def areEquivalent(num 1,num 2): | return divSum(num 1) == divSum(num 2);`

Note: The provided answer correctly defines a function to calculate the sum of divisors for a given number and another function to compare the sums of divisors for two numbers, thereby fulfilling the requirement of checking whether the sums are equivalent.

12. **Answer:** `def set_middle_bits(n): | return (n >> 1) ^ toggle_middle_bits(n) : |return 1|return n^set_middle_bits(n)`

Note: The provided answer correctly defines two functions, where 'set_middle_bits' shifts the input number

End of Answer Key